

	$= 4440 \text{ W}$
4(a)	Nuclear fusion is the combination of two smaller nuclei to form a larger nucleus, with a release of energy.
4(b)(i)	change in mass, $\Delta m = 4(1.007293) - 4.001572 = 0.0276u$ energy released $= \Delta m c^2 = (0.0276)(1.66 \times 10^{-27})(3.00 \times 10^8)^2$ $= 4.12 \times 10^{-12} \text{ J}$
4(b)(ii)	In 1 s, energy emitted by the Sun, $E = 3.85 \times 10^{26} \text{ J}$

No.	Solution
	$\text{loss of mass, } m = \frac{E}{c^2} = \frac{3.85 \times 10^{26}}{(3.00 \times 10^8)^2} = 4.28 \times 10^9 \text{ kg}$ Average rate of loss of mass = $4.28 \times 10^9 \text{ kg s}^{-1}$
4(c)(i)	$\text{Time} = \frac{1.99 \times 10^{30}}{4.28 \times 10^9} = 4.652 \times 10^{20} \text{ s}$ $= \frac{4.652 \times 10^{20}}{(3600)(24)(365)} = 1.48 \times 10^{13} \text{ years}$
4(c)(ii)	The assumption that the rate of loss of mass being constant is not valid. As more mass is lost, the remaining mass of the Sun would be less, and the rate of fusion would correspondingly be less.
4(d)(i)	The gravitational force on the planet of mass m by the Sun of mass M provides its centripetal force for its circular orbit of radius r about the Sun, i.e. $\frac{GMm}{r^2} = mr\omega^2 = mr \left(\frac{2\pi}{T} \right)^2 = \frac{4\pi^2 mr}{T^2}$ $T^2 = \left(\frac{GM}{4\pi^2} \right) r^3 = kr^3$ where $k = \frac{GM}{4\pi^2}$ since M is taken as constant.
4(d)(ii)	Using the relation in (d)(i), orbit period of Earth T_E is related to the orbit period of Jupiter T_J by $\frac{T_E^2}{T_J^2} = \frac{1.50^3}{7.79^3} = 0.007139$ $\frac{T_E}{T_J} = \sqrt{0.007139} = 0.08449$ $T_J = 11.8 T_E$ Speed of planet, $v = r\omega = \frac{2\pi r}{T} \rightarrow v^3 \propto \frac{r^3}{T^3} \propto \frac{T^2}{T^3} \propto \frac{1}{T}$ $\frac{v_E}{v_J} = \left(\frac{T_J}{T_E} \right)^{1/3} = (11.8)^{1/3} = 2.28$ $v_E = 2.28 v_J$
5(a)(i)	The velocity is obtained from the gradient of the tangent of the graph at 0.60 s